

8 HYDRODYNAMIC COMPONENTS

These cards are required for NEW type problems and may be entered for RESTART problems. Hydrodynamic systems are described in a NEW problem. In a RESTART problem, the hydrodynamic systems may be modified by deleting, adding, or replacing components. The resultant problem must describe at least two volumes and one junction. The hydrodynamic card numbers are divided into fields, where CCC is the component number (the component numbers need not be consecutive), XX is the card type, and NN is the card number within type. When a range is indicated, the numbers need not be consecutive.

For junctions, an old format has been used in the past to connect volumes. In the old format, CCC000000 was used if the connection was to the inlet side of the component and CCC010000 was used if the connection was to the outlet side of the component. The use of this format is discouraged and the expanded connection format is preferred because it is more general and easier to understand. In the expanded format, the connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. A nonzero N specifies the expanded format. The number N equal to 1 and 2 specifies the inlet and outlet faces, respectively, for the volume's coordinate direction (see Section 3.1.1.8 of Volume I). The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For example, in a 5 volume pipe component, the old format inlet connection code of CCC000000 is equivalent to CCC010001 in the expanded format, and the old outlet connection code of CCC010000 is equivalent to CCC050002 in the expanded format. In the following component descriptions, only the expanded format connection code will be used.

8.1 General Information

8.1.1 Component Name and Type, CCC0000

This card is required for each component.

- | | |
|-------|---|
| W1(A) | Component name. Use a name descriptive of the component's use in system. A limit of 8 characters is allowed for most computers, e.g., workstations, CRAY, and IBM computers. |
| W2(A) | Component type. Enter one of the following component types, SNGLVOL, TMDPVOL, SNGLJUN, TMDPJUN, PIPE, ANNULUS, PRIZER, CANCHAN, BRANCH, SEPARATR, JETMIXER, TURBINE, ECCMIX, VALVE, PUMP, CIRCLTR, MTPLJUN, ACCUM, SDBVOL, MULTID or the command DELETE. The command DELETE is allowed only in RESTART problems, and the component number must be an existing component at the time of restart. The DELETE command deletes the component. |

The remaining cards for each component depend on the type of component.